DAY 4 HOMEWORK

Plan Your First AI Application Before Writing Python

Name:Sohan		**Date:**	
Project/App Name:	AIpad		

Assignment: Complete all eight planning items for your own AI capstone idea. Do not begin coding until your plan is specific, testable, and easy to explain.

Submission: Submit this completed document and upload it to your GitHub project folder using a clear filename such as day4_ai_app_plan.docx.

1. Clear Problem Statement

Write one specific sentence using this structure: "Many [users] struggle with [specific problem] because [reason]."

Helpful reminder: Describe the problem, not the features of your app. Avoid broad words such as "people," "things," or "everything."

My problem statement:

Many coders and developers struggle with having a place to put their code and write new code easily and paste it into their application, because notepads and coding apps are 2 seperate apps and they need to be switching between both to store short pieces of code. This AIpad lets coders write new pieces of code easily and separately and it also has AI syntax correction and also has text correction. You do not need to open a new project to create pieces of code.

2. User Story

Complete: "As a [type of user], I want to [action], so that [benefit]."

My user story:

As a coder, I want to easily write and store new blocks of code while also having AI check it, so that I can save those pieces of code for when I need them so I can experiment with different blocks in my projects without having to open a new file to create it.

3. Input–Process–Output (IPO) Chart

Input is what the user gives the app. **Process** is what the app does. **Output** is what the user receives.

INPUT	PROCESS	OUTPUT
Type code, then click AI fix.	Code is sent to the AI and then is sent back with good syntax.	Old code gets replaced by new code.
Type text, then click AI fix.	Text is sent to the AI and then is sent back with good grammar and spelling..	Old text gets replaced by new text
Click AI generate, then type what you want to generate.	Prompt is sent to AI, then the AI sends the generated content back.	Content appears on the notepad.

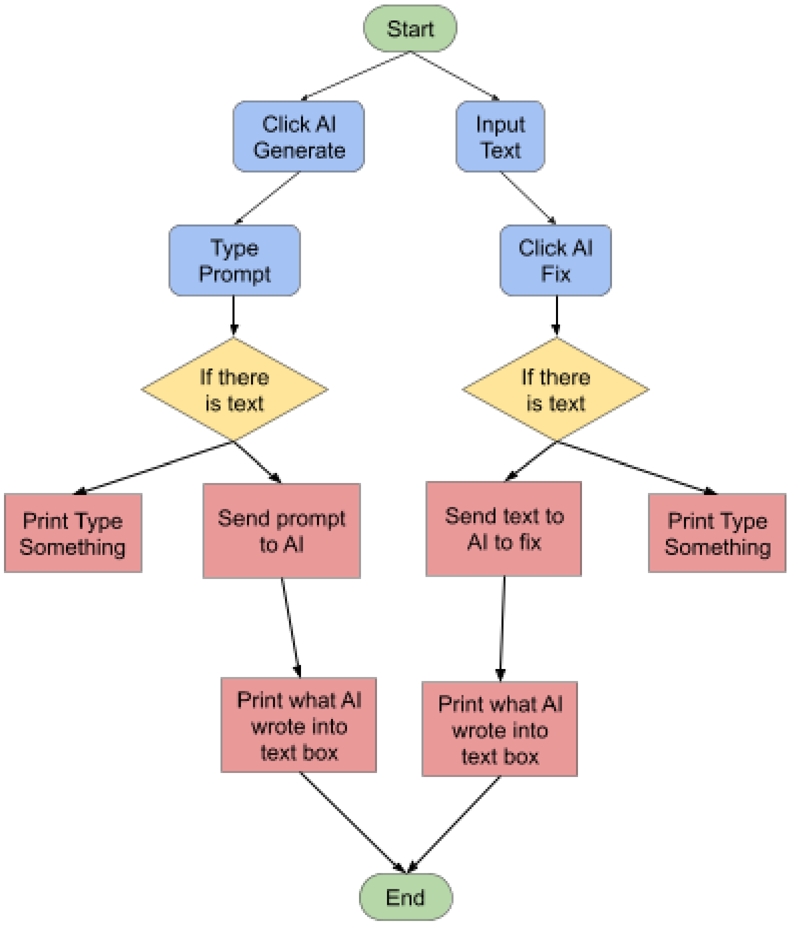
4. Pseudocode

Write your program plan in plain English. Include: START, user input, validation, processing, output, and END. Use IF/ELSE when the app must make a decision.

1	User types code or text.
2	If User clicks AI Fix
3	Then check if there is anything typed.
4	If there is content
5	Then send to AI and type fix(the app types fix at end)
6	Else, say “Type Something” and don't let user click fix
7	If AI sends back
8	Then replace existing content with new content
9	Display new content
10	If user clicks AI generate
11	Then open a text box that says Type What you want to generate.
12	If user clicks send button
13	Check If there is something typed
14	If yes
15	Then send text to AI
16	Else, say type something
15	Then display whatever AI wrote in AIpad

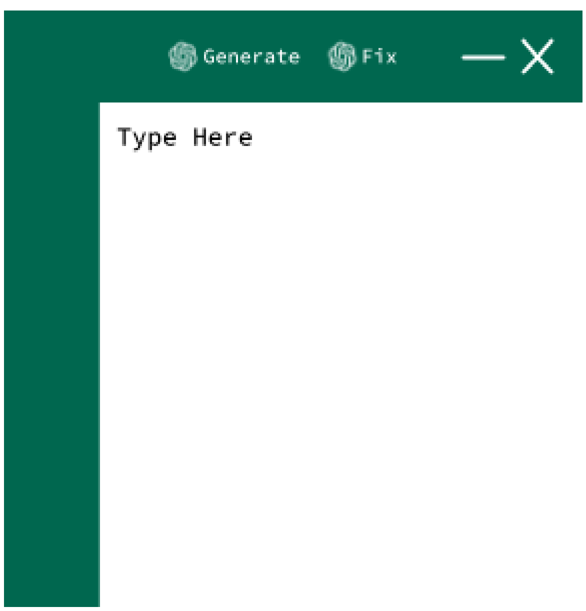
5. Flowchart

Draw the order of your app. Include at least: Start, Input, one Decision, Process, Output, and End. Label each arrow clearly.



6. User Interface Sketch

Sketch what the user will see. Include: app title, input areas, button(s), warning/error location, and output area.



7. Two Test Cases

For each test, show exactly what the user enters and what the app should return. Use one normal test and one edge case, such as empty input, unclear wording, or missing information.

Test Case 1

Test purpose	To see if AI returns text back with good spelling.
Exact input	Bad spelling and grammar. I am preparing to wented to the park tomorrow.
Expected output	Text with good spelling and grammar.
How I will know it passed	If the text has good spelling and grammar.

Test Case 2

Test purpose	To see if AI returns code back with good syntax.
Exact input	Code with syntax errors.
Expected output	Fixed code.
How I will know it passed	If the code is fixed.

8. Scope Statement: Version 1

Keep your first version small. List only the most important features under “Will Do.” Put future ideas and complicated integrations under “Will Not Do Yet.”

VERSION 1 WILL DO	VERSION 1 WILL NOT DO YET
AI correction	Tabs
Storing text	Autocomplete
Copy paste	

One-Breath Project Explanation

Explain your project in 2–3 sentences: who it helps, what information it uses, what it does, and what result it provides.

My app helps coders and regular people write by offering a simple space to write and store code and text for later.

Submission Checklist

- ☐ I completed all eight sections.
- ☐ My problem statement is specific.
- ☐ My input, process, and output match each other.
- ☐ My pseudocode and flowchart show the same sequence.
- ☐ My interface sketch includes input, button, messages, and output.
- ☐ My two test cases include exact input and expected output.
- ☐ My Version 1 scope is small enough to build.
- ☐ I reviewed spelling and clarity.
- ☐ I saved the file with a clear lowercase filename.
- ☐ I uploaded the completed document to GitHub.

Teacher Review (40 points)

Required Item	Points	Teacher Notes
1. Problem statement	/ 5	
2. User story	/ 5	
3. IPO chart	/ 5	
4. Pseudocode	/ 5	
5. Flowchart	/ 5	
6. Interface sketch	/ 5	
7. Two test cases	/ 5	
8. Scope statement	/ 5	